6. Data Dictionary

This dictionary is based on column names and observed values in the supplied CSV. Where a field's business definition is not provided, the description is explicitly provisional.

Field | Description | Inferred type | Missing values | Distinct non-null values

`vehicle_id` | Vehicle record identifier | Categorical/text | 0 (0.00%) | 100000

`model_name` | Vehicle model label | Categorical/text | 0 (0.00%) | 17919

`manufacturer` | Manufacturer label | Categorical/text | 0 (0.00%) | 200

`model_year` | Model year (2018–2026 in this file) | Integer | 0 (0.00%) | 9

`manufacturing_country` | Manufacturing country label | Categorical/text | 0 (0.00%) | 10

`market_region` | Market region label | Categorical/text | 0 (0.00%) | 7

`category` | Vehicle category | Categorical/text | 0 (0.00%) | 15

`battery_capacity_kwh` | Battery capacity, kWh | Numeric | 0 (0.00%) | 21832

`battery_type` | Battery chemistry/type label | Categorical/text | 0 (0.00%) | 5

`battery_cycle_life` | Battery cycle-life value (source unit/definition not provided) | Numeric | 2,028 (2.03%) | 3338

`battery_degradation_pct_5yr` | Five-year battery degradation percentage | Numeric | 0 (0.00%) | 2687

`range_km` | Range, km | Numeric | 0 (0.00%) | 6020

`efficiency_wh_per_km` | Energy efficiency, Wh/km | Numeric | 0 (0.00%) | 38359

`motor_power_kw` | Motor power, kW | Numeric | 0 (0.00%) | 6383

`torque_nm` | Torque, Nm | Numeric | 0 (0.00%) | 18613

`num_motors` | Number of motors | Integer | 0 (0.00%) | 3

`drivetrain` | Drivetrain label | Categorical/text | 0 (0.00%) | 3

`top_speed_kmph` | Top speed, km/h | Numeric | 0 (0.00%) | 2894

`acceleration_0_100_sec` | 0–100 km/h acceleration time, seconds | Numeric | 0 (0.00%) | 2790

`max_dc_charging_kw` | Maximum DC charging power, kW | Numeric | 0 (0.00%) | 3471

`charging_dc_fast_time_min` | DC fast-charging time, minutes (exact charge window not specified) | Numeric | 0 (0.00%) | 485

`charging_ac_time_hr` | AC charging time, hours (exact charge window not specified) | Numeric | 0 (0.00%) | 2969

`num_charging_ports` | Number of charging ports | Integer | 0 (0.00%) | 2

`num_seats` | Seat count | Integer | 0 (0.00%) | 46

`weight_kg` | Vehicle weight, kg | Numeric | 0 (0.00%) | 20077

`cargo_capacity_liters` | Cargo capacity, litres | Numeric | 0 (0.00%) | 11424

`price_usd` | Price, USD | Numeric | 0 (0.00%) | 99430

`insurance_cost_usd_yearly` | Annual insurance cost, USD/year | Numeric | 0 (0.00%) | 85266

`warranty_years` | Warranty duration, years | Integer | 0 (0.00%) | 7

`resale_value_pct_3yr` | Resale value after three years, percent | Numeric | 3,057 (3.06%) | 6625

`safety_rating_stars` | Safety rating, stars (observed 1–5) | Integer | 0 (0.00%) | 5

`autonomous_level` | Autonomous level (observed 0–5; scale source not supplied) | Integer | 0 (0.00%) | 6

`infotainment_screen_inch` | Infotainment screen size, inches | Numeric | 1,506 (1.51%) | 141

`co2_saved_tons_per_year` | CO■ saved per year, metric tons (calculation basis not supplied) | Numeric | 0 (0.00%) | 10326

`annual_sales_units` | Annual sales units (scope/measurement basis not supplied) | Numeric | 989 (0.99%) | 24233

`customer_rating` | Customer rating (scale not explicitly documented; observed 1.75–5) | Numeric | 1,931 (1.93%) | 298

`recommended_use_case` | Recommended use case label | Categorical/text | 0 (0.00%) | 7

Fields requiring source confirmation

- `battery_cycle_life`: confirm cycle-test conditions and end-of-life threshold.
- `charging_dc_fast_time_min` and `charging_ac_time_hr`: confirm start/end state of charge and charger assumptions.
- `autonomous_level`: confirm the standard used for levels 0–5.
- `co2_saved_tons_per_year`: confirm baseline vehicle, annual distance, grid mix, and calculation method.
- `annual_sales_units`: confirm geographic and time scope, and whether values are observed or generated.
- `customer_rating`: confirm source, respondent base, and scale.
- `resale_value_pct_3yr`: confirm valuation methodology and reference price.